

Two-stage process analysis using the process-based performance measurement framework and business process simulation

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ABSTRACT

Many enterprises have recently been pursuing process innovation or improvement to attain their performance goals. To align a business process with enterprise performances, this study proposes a two-stage process analysis for process (re)design that combines the process-based performance measurement framework (PPMF) and business process simulation (BPS). The two-stage analysis consists of macro and micro analyses of business processes. At the early stage of business process analysis (BPA), macro process analysis is conducted to identify the influence of a business process on a target key performance indicator (KPI) or the contribution of a target KPI to other KPIs. If target business processes that need improvement are identified through the macro process analysis and to-be processes are newly designed, micro process analysis using simulation is conducted to predict the performance. The proposed method is validated by application to a real business process within the setting of a large Korean company.

By using the proposed, two-stage process analysis, company staff involved in process innovation projects can determine the processes with the greatest influence on enterprise strategy, and can systematically evaluate the performance prediction of the newly designed process.

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1. Introduction

In today's dynamic business environment, the ability to improve performance is a critical requirement for business organizations. Therefore, many enterprises have recently been pursuing business process innovation or reengineering to attain their performance goals. A business process is defined as a sequence of activities to achieve a complete business goal. To comprehensively support business process execution, the concept of business process management (BPM) has been proposed. BPM is the identification, understanding, and management of business processes linked with people and systems and across organizations (Debevoise, 2005).

For aligning with enterprise strategies, BPM activities must be closely related to performance metrics because the metrics are the drivers and evaluators of business processes. Since the metrics indicate how well the business is doing relative to a defined strategy, they help managers to derive better business decisions. Therefore, for effective BPM, it is imperative to establish the relationship among enterprise strategies, business processes and metrics, before deploying the BPM system.

The BPM cycle is composed of diagnosis, process design, system configuration, and process enactment (Van der Aalst, Ter Hofstede, & Weske, 2003). In the diagnosis phase, the operational processes are analyzed to identify problems and find areas for improvement. In the process design phase, to-be processes are newly designed and analyzed. According to Van der Aalst, Weske, and Wirtz (2003), there are basically three types of business process analysis (BPA): validation, verification and performance analysis. However, prior to these detailed process analyses, macro process analysis must be performed to determine which business processes are linked or aligned to specific enterprise strategies to attain the business goal. In other words, for process innovation or improvement, the most significant process to produce a successful performance should be identified, as well as the performance indicators which are most significantly influenced when a specific process is executed.

After the macro process analysis, performance measurement is needed for micro process analysis, and metrics should be established for performance measurement. Van der Aalst and Van Hee (2002) differentiated between the analysis of the qualitative and quantitative aspects of business workflows. The former mainly concerns the logical correctness of the defined process and can be analyzed typically by Petri nets, whereas the latter mainly concerns the performance of the defined process, and can be analyzed

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typically by simulation method. Simulation is a widely used effective tool for detailed BPA, due to its ability to evaluate the impact of process changes and new processes in a model environment through the creation of what-if scenarios. Simulation is also promoted to enable the examination and testing of decisions prior to actually making them in the real environment (April, Better, Glover, Kelly, & Laguna, 2005).

In this paper, we integrate the two aforementioned analysis methods, and propose a layered approach to analyze the effectiveness and efficiency of business processes. The proposed method is a two-stage process analysis for process (re)design using the process-based performance measurement framework (PPMF) and business process simulation (BPS). This two-stage analysis consists of macro and micro analyses of business processes. At the early stage of BPA, macro process analysis is conducted to identify the influence of a business process on a target key performance indicator (KPI) or the contribution of a specific KPI to other KPIs. If target business processes that need improvement are identified through the macro process analysis, micro process analysis using simulation is conducted for performance evaluation at a detailed level. The macro and micro process analyses in this paper are based on the well-established PPMF in which the relationship of business processes and enterprise KPIs is clearly defined and managed (Han & Kang, 2007).

The rest of this paper is organized as follows. Section 2 summarizes the related works. Section 3 describes the overall PPMF structure. Section 4 presents the procedure of the two-stage process analysis. Section 5 presents a case study to demonstrate the applicability of two-stage process analysis. Finally, the last section summarizes the results and suggests directions for further research.

2. Related works

Related works can be categorized into those investigating performance measurement, process analysis, and process simulation. Among the several research works about the performance measurement of an enterprise, Ghalayini, Noble, and Crowe (1997) classified the performance measures for a manufacturing enterprise into the following three layers: general area of success, performance measure, and performance indicator. They defined the hierarchical relationships between the layers, but did not show the relationship between a performance measure and business process. Berler, Pavlopoulos, and Koutsouris (2005) classified a hospital's KPIs based on the concept of balanced scorecard (BSC). They also defined hospital workflows in detail, and then assigned a KPI to each unit task-level activity. However, they addressed the interrelationship between KPI and a business process merely by simple one-to-one mapping at a single flat level. Chan and Qi (2003) and Beretta (2002) decomposed the business process hierarchically into sub-processes, and then proposed a method of calculating the performance measure by summing up the measures of the lowest level activities. Theirs is a top-down approach that defined the business processes level-by-level, and mapped the performance measure to each business process. Nevertheless, they suggested that the business process has a simple one-to-one relationship with a performance measure. Bititci (1995) conducted a case study that applied the information systems modeling techniques to the modeling of a performance measurement system and implied that KPIs are interrelated with multiple and weighted relationships. Munehira and Mori (2006) suggested that it is necessary to identify which processes should be improved or created to achieve the business performance goals described in BSC. For this purpose, they defined the relations between BSC structure and business model elements, yet they did not implement their proposed method to a real business case.

Since Petri nets can model the dynamic behavior of business processes based on a mathematical formalism, they are widely used for BPA (Feng & Jiang, 2006; Salimifard & Wright, 2001; Van der Aalst, 1999; Van der Aalst & Ter Hofstede, 2000). Among the various Petri net techniques, reachability tree analysis methods were mainly proposed for business workflow analysis because they can detect erroneous states of the workflow such as deadlock and livelock, and can trace the state transitions (Choi, Zhao, & Han, 2006; Van der Aalst & Van Hee, 2002). However, these researches emphasized the detailed qualitative and quantitative analyses to investigate the sensitivity of process efficiency.

In the area of BPS, April et al. (2005) suggested the usefulness of simulation for the business process reengineering (BPR) by presenting the case study of a personal claims process at an insurance company. Mevius and Oberweis (2005) presented that Petri-net based simulation is useful for the evaluation of process performance. Greasley (2003) demonstrated the ability of BPS to incorporate system variability, scenario analysis and a useful display to communicate process performance for BPR. An and Jeng (2005) presented the development of a system dynamics model of business processes for the accurate description of system behavior and what-if analysis. However, their model focused on detailed process behavior having a single flat level process hierarchy.

3. Process-based performance measurement framework (PPMF)

This section describes PPMF. Fig. 1 shows the overall structure of PPMF. PPMF consists of three sub-models: KPI model, process model, and K-P model. The first two models define the KPIs and processes, respectively, while the latter describes the relationship between the first two. These three models are closely related to each other and form an integrated model for performance measurement. The remainder of this section explains each model in detail.

The KPI model, depicted in the lower left part of Fig. 1, defines KPIs and their correlation. In the model, KPIs are hierarchically classified into three groups according to the managerial decision making levels: strategic level KPI (SLK), tactical level KPI (TLK), and operational level KPI (OLK). As a strategic plan is elaborated by tactical plans and a tactical plan is decomposed into operational plans in an enterprise, SLKs and TLKs are supported by TLKs and OLKs, respectively. To define the relationships between SLKs and TLKs and those between TLKs and OLKs, the contribution index is introduced. The contribution index is a measure of the contribution of a specific KPI to another or higher level KPI. It is measured with a 3-point scale: minor contribution, normal contribution, and major contribution. The right part of Fig. 2, which is an example of an integrated view of a PPMF, shows an example of the KPI model represented as a matrix form. The KPI model in PPMF has a whole list of all the KPIs in an enterprise. When two KPIs are related to each other, the related cell is marked by contribution index. In the figure, the numbers 1, 2 and 3 indicate 'minor contribution', 'normal contribution', and 'major contribution', respectively. For example, 'internal process lead-time' SLK contributes to 'productivity' SLK by normal scale (2).

The second part in PPMF is the process model. As depicted in the lower right part of Fig. 1, business processes are also classified into three levels in accordance with the size of the control span: enterprise level, process level, and sub-process level. Fig. 3 shows the hierarchy of the process model. A process in a sub-process level consists of a network of activities, and the process simulation for the detail-level analysis is conducted at this level. The definition of process, sub-process and activity in this paper is compliant with the definition of workflow management coalition (WfMC) (Workflow Management Coalition, 1999). In Fig. 2, the upper left part

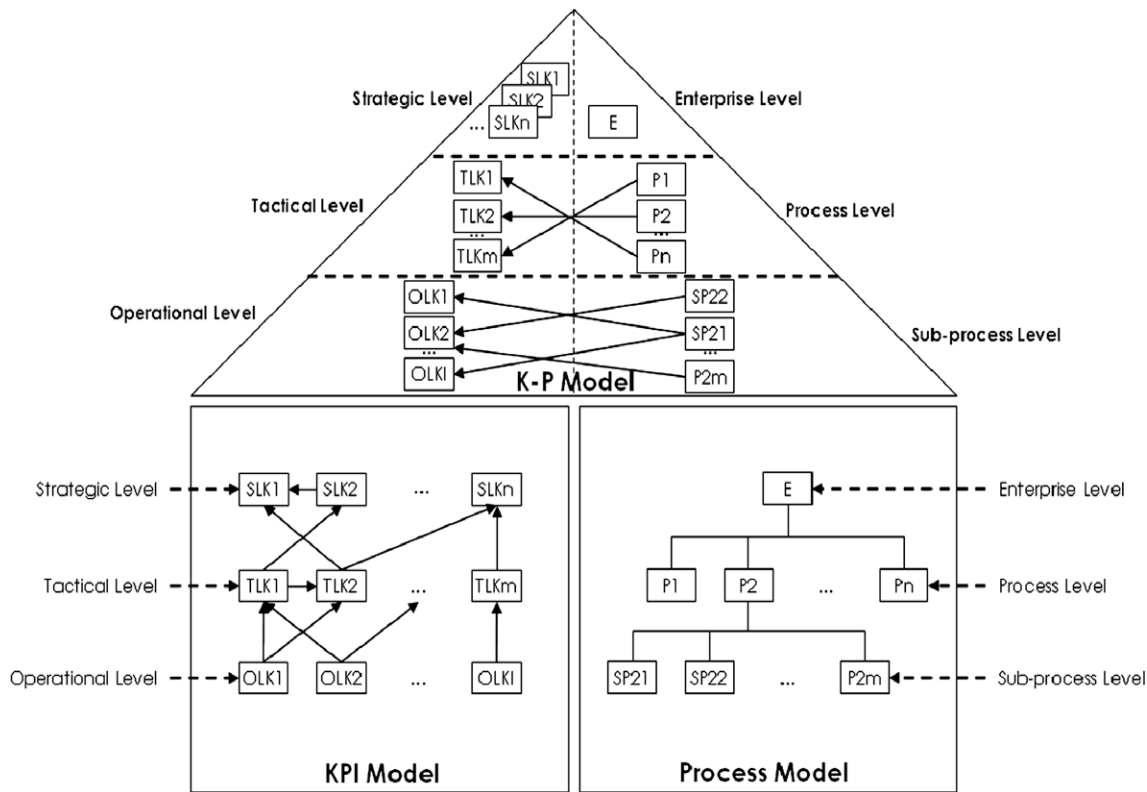


Fig. 1. PPMF structure.

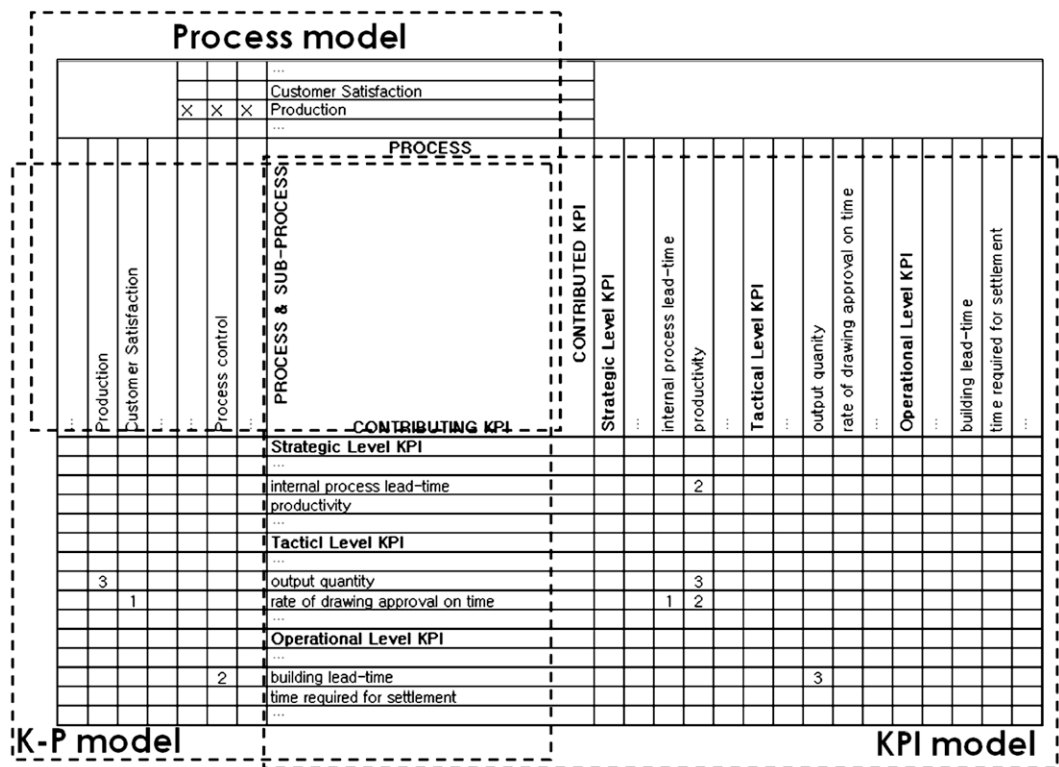


Fig. 2. Integrated view of PPMF (simplified).

shows an example of a process model which describes the relationship among processes and sub-processes. For example, in the figure, 'process control' is a sub-process of 'production' process.

Finally, the K-P model, depicted in the upper part of Fig. 1, represents the relation between KPI and business processes. Enterprise, process, and sub-process levels in the process model

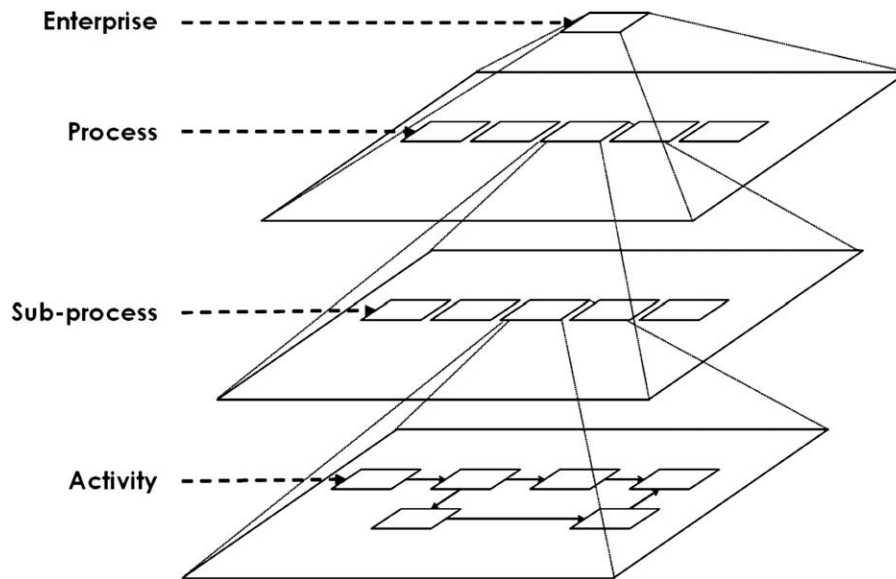


Fig. 3. Hierarchy of the process model.

correspond to SLK, TLK, and OLK in the KPI model, respectively. To show the degree of alignment between KPI and business processes, the influence index, which is a measure of the influence of a business process on a specific KPI, is determined in the K–P model. The influence index is measured with the following 3-point scale for the business process on TLKs and the sub-process on OLKs: minor influence, normal influence, and major influence. The lower left part of Fig. 2 shows the K–P model which describes the influence indexes. In the figure, the numbers 1, 2 and 3 indicate 'minor influence', 'normal influence', and 'major influence', respectively. For example, 'Production' process has a major influence on 'output quantity' TLK. Note that the contribution and influence indexes in PPMF are determined by the Delphi method of an expert group who are the representatives of each department. The procedure to develop a PPMF in an organization has been previously addressed by the authors in Han and Kang (2007).

4. Two-stage process analysis

This section explains the two-stage process analysis. The proposed BPA consists of two stages: macro and micro process analyses. The former focuses on the effectiveness of the process. The business processes with the greatest influence on a target SLK are selected based on PPMF at this stage. Afterwards, the performances of the selected processes are reviewed, and the to-be processes are newly designed. The latter concentrates on the to-be processes to investigate process efficiency. The performance predictions of new processes are made by the process simulation, and evaluated at this stage.

4.1. Macro process analysis

What-if analysis is performed based on PPMF during the macro process analysis stage to determine which processes should be executed successfully to achieve a target SLK. First of all, the TLKs with the greatest contribution to the target SLK in the KPI model are identified, and then the business processes that directly influence the target TLK in the K–P model can be selected as candidate business processes. If no process directly influences the target TLK, we alternatively use a target OLK with the greatest contribution to the target TLK in the KPI model, and then identify a sub-process

that directly influences the selected target OLK in the K–P model. Then the higher level process of the identified sub-process can be selected as a candidate process in the process model.

The use of PPMF enables another problem to be addressed: determining which SLKs are influenced if a target process or sub-process is innovated or improved. To solve this problem, the TLK that is influenced by the target process in the K–P model should be identified, after which the SLKs that are contributed by the selected TLK in the KPI model can be determined. At the sub-process level, we also determine firstly the OLK that is influenced by the target sub-process in the K–P model, secondly the TLK that is contributed by the selected OLK, and thirdly the SLKs that are affected by the selected TLK in the KPI model.

4.2. Micro process analysis

After the macro process analysis, micro process analysis is performed. Micro process analysis, which is a kind of quantitative BPA using a simulation tool, focuses on evaluating the ability to meet performance requirements with respect to throughput times, service levels, and resource utilizations, etc. OLKs are used as metrics for micro process analysis, whereas SLKs and TLKs are major concerns of the macro process analysis stage.

After the target business process is selected for innovation or improvement, the performance of the as-is process is reviewed, after which a to-be process is newly designed. At the micro process analysis stage, the process simulation for performance prediction of the newly designed process is conducted as follows. Firstly, a process model is built by using simulation software. Next, simulation parameters such as processing time and other experimental data for each activity of the business process are prepared, after which process simulation is performed. During the simulation execution, various performance metrics are measured and compared with the actual performances of current process for evaluating the to-be process.

5. Case study: Company S in Korea

This section presents the result of a case study of *Company S* in Korea in order to show the applicability of the proposed, two-stage process analysis. *Company S*, a large ship builder, is one of the

biggest companies in Korea's shipbuilding industry with about 7000 employees.

5.1. Case Study: macro process analysis

For the macro process analysis, we developed a PPMF of Company S, shown in part in Fig. 4. In the figure, the arrow shows

the macro process analysis procedure. In the case study, we focus on the 'productivity' SLK. To determine a process and sub-process that should be improved for increasing the 'productivity' SLK, we firstly identify that the major contributing TLK to the 'productivity' SLK is 'output quantity' in the KPI model (arrow number ① in Fig. 4) from PPMF. Then we can determine that the major influencing process on the 'output quantity' TLK is

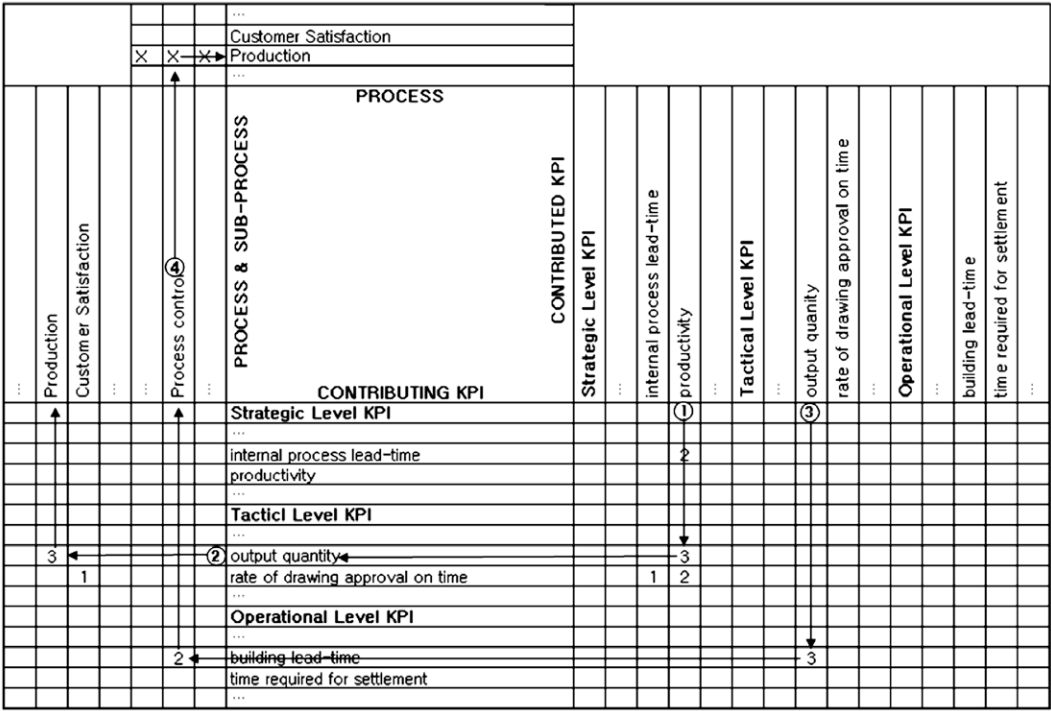


Fig. 4. Macro process analysis based on PPMF of Company S.

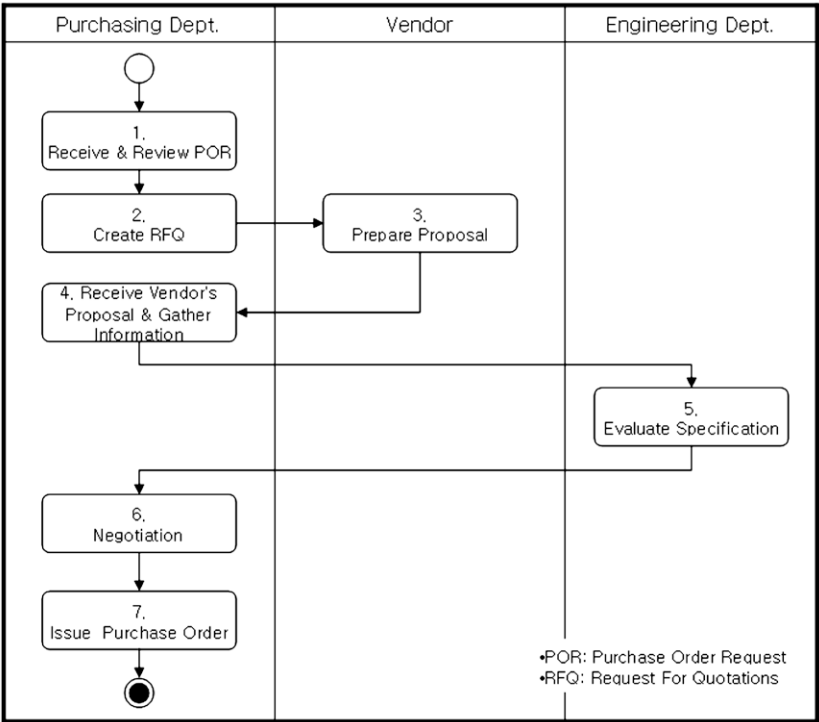


Fig. 5. To-be process of 'placing purchase order'.

'production' process in the K-P model (arrow number ② in Fig. 4).

To investigate business processes in more detail, we identify the major influencing sub-process on the 'output quantity' TLK. The major contributing OLK to the 'output quantity' TLK is the 'building lead-time' OLK in the KPI model (arrow number ③ in Fig. 4). This result indicates that the major influencing sub-process on the 'building lead-time' OLK is 'process control' in the K-P model. Finally, we can determine that 'process control' is the sub-process of 'production' process in the process model (arrow number ④ in Fig. 4).

5.2. Case study: micro process analysis

In the micro process analysis, we addressed the problem of which sub-process we have to improve for reducing the 'cost' SLK. Macro process analysis for this problem is conducted as follows. Firstly, the 'inventory cost' is identified as a major contributing TLK to the 'cost' SLK. Secondly, the 'purchasing' process is

determined as the process with the greatest influence on the 'inventory cost' TLK. Finally, among the sub-processes of 'purchasing' process, the 'placing purchase order' sub-process is selected as a target process to reduce the 'cost' target SLK.

After the review of the current performances of the existing 'placing purchase order' sub-process, the to-be process is redesigned, as shown in Fig. 5. This sub-process consists of a sequence of seven activities. (1) Receive & Review POR (Purchase Order Request): The purchasing department receives PORs from various departments and reviews them. (2) Create RFQ (Request for Quotations): The purchasing department creates an RFQ, and sends it to a vendor. (3) Prepare Proposal: The vendor reviews the RFQ, then prepares a proposal, and sends it to the purchasing department. (4) Receive the Vendor's Proposal & Gather Information: The purchasing department receives the proposal, gathers related information, prepares a specification, and requests the engineering department to evaluate it. (5) Evaluate Specification: The engineering department evaluates the specification, and sends the evalua-

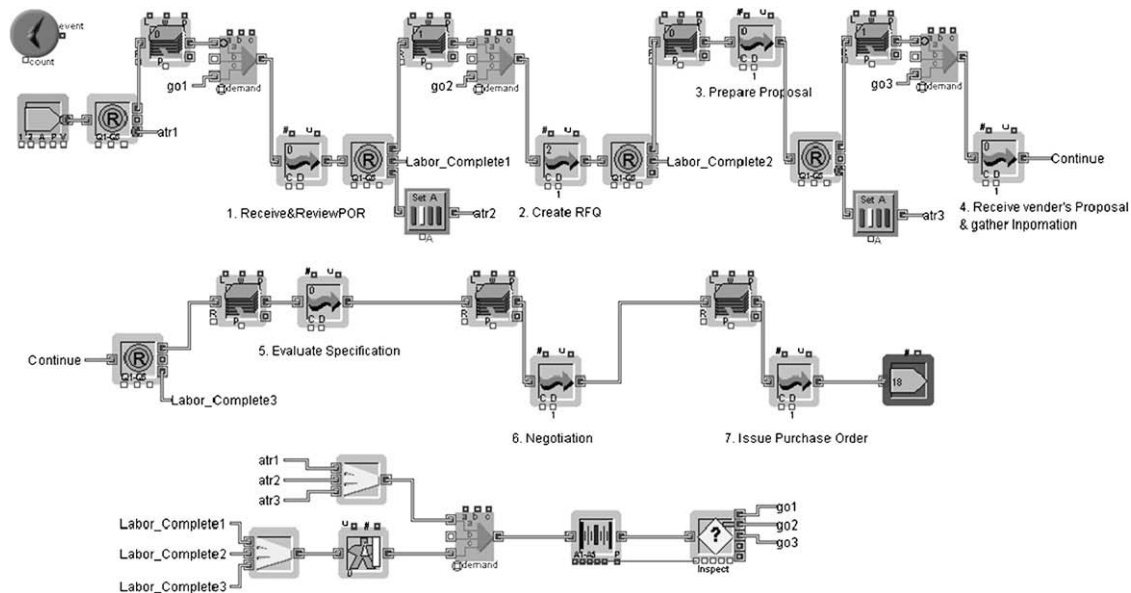


Fig. 6. Simulation model using ExtendSim.

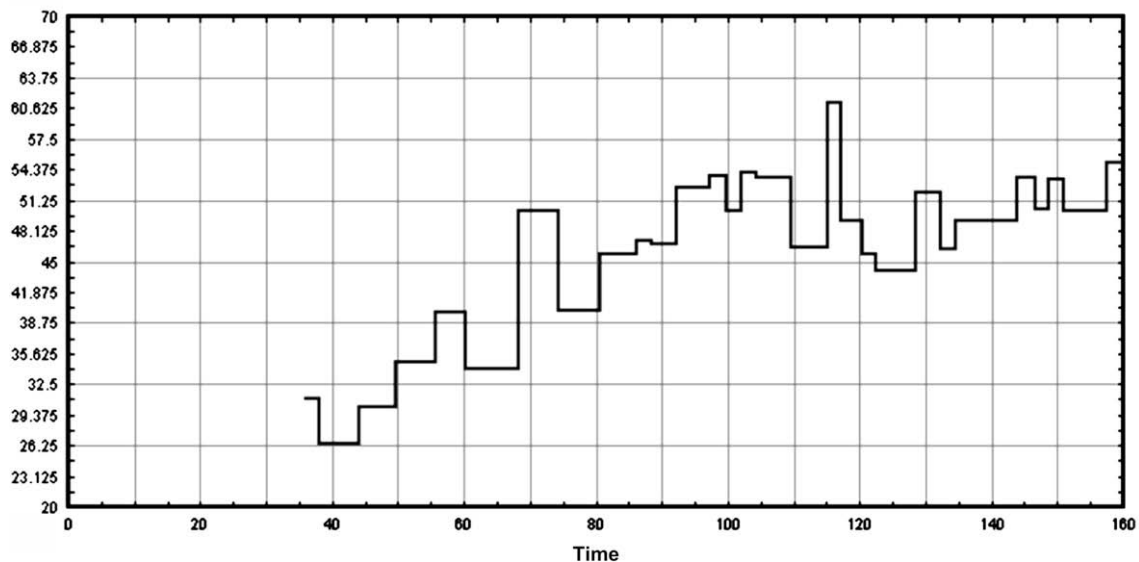


Fig. 7. POR throughput time trend.

tion result to the purchasing department. (6) Negotiation: The purchasing department negotiates with the vendor. (7) Issue Purchase Order: After negotiation, the purchasing department issues a purchase order to the vendor. Activities 1, 2, and 4 in Fig. 5 are executed by a labor pool comprised of 4 workers, while activities 6 and 7 have a dedicated worker. Note that the resource models for the vendor and the engineering department are ignored.

For the performance prediction of the new process, a simulation model, shown in Fig. 6, is built using ExtendSim tool (Imagine That Inc., 2007). One month period (160 h) is simulated and various performance metrics are measured. The average throughput time for each POR from various departments is 46.4 h. The trend of POR throughput time is shown in Fig. 7. The average resource utilization of the labor pool is 88%. By using the process simulation method, we can predict the performance of the newly designed process and easily compare it with that of the as-is process. The results indicate that the average throughput time is reduced by 16% and the average resource utilization of the labor pool is enhanced by 18% in the new process.

6. Conclusions

Well-designed process analysis supports the evaluation of business processes and the improvement of current process practices. Process analysis linking business processes to enterprise strategy requires a layered approach to analyze the effectiveness and efficiency of business processes. This paper proposes a two-stage process analysis comprised of macro and micro process analyses. For aligning enterprise performances with business processes, the structure of PPMF is also presented.

By using the proposed, two-stage process analysis, company staff involved in the process innovation projects can determine the processes with the greatest influence on enterprise strategy, and can systematically evaluate the performance prediction of the newly designed processes. In other words, the business process with the greatest influence on a specific SLK is selected at the macro process analysis stage by using PPMF. Afterwards, the performance of the as-is process is reviewed, and a to-be process is newly designed. The performance of the new process is predicted by using process simulation at the micro process analysis stage.

Further research on the implementation of a tool to support our approach is required. The implementation of the current PPMF is a stand-alone system since it is built on an Excel sheet and is thus not integrated with any commercial simulation tool. Therefore, further research should focus on developing an integrated BPA tool that supports PPMF.

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